

CBSE Class 12 Pe — Training in Sports

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

- Q1.** The Swedish word "Speed Play" is also known as: (A) Fartlek training (B) Interval method (C) Continuous method (D) Pace method [PYQ 2024] | Confidence: Very High
- Q2.** In which type of exercises can the movement NOT be seen directly? (A) Iso-tonic (B) Iso-kinetic (C) Iso-metric (D) Plyometric [PYQ 2024] | Confidence: Very High
- Q3.** In the word 'Isometric', 'Iso' means constant and 'metric' means: (A) Tension (B) Force (C) Length (D) Contraction [PYQ 2024] | Confidence: Very High
- Q4.** Which of the following training methods was developed by Gösta Holmér? (A) Interval Training (B) Fartlek Training (C) Continuous Training (D) Circuit Training [PYQ] | Confidence: High
- Q5.** The Interval Training method is primarily based on the principle of: (A) Overload (B) Effort (C) Effort and Recovery (D) Continuous tension [PYQ] | Confidence: Very High
- Q6.** The **1500m** run in physical fitness selection criteria is primarily conducted to find the athlete's: (A) Endurance ability (B) Speed (C) Strength (D) Explosive power [PYQ] | Confidence: High
- Q7.** Which of the following is NOT a type of coordinative ability? (A) Orientation Ability (B) Acceleration Ability (C) Coupling Ability (D) Differentiation Ability [PYQ 2025] | Confidence: Very High
- Q8.** The sports training cycle that typically lasts for a medium duration, ranging from **3** to **6** weeks, is termed as a: (A) Micro cycle (B) Meso cycle (C) Macro cycle (D) Olympic cycle [Practice] | Confidence: Very High
- Q9.** The longest training cycle, which usually lasts from **3** to **12** months and aims at peak performance during the main competition, is the: (A) Micro cycle (B) Meso cycle (C) Macro cycle (D) Mini cycle [Practice] | Confidence: Very High
- Q10.** The 'Isokinetic' method of muscle training, involving specially designed machines to apply maximal force throughout the range of motion, was introduced by: (A) J.J. Perrine (B) Morgan and Adamson (C) Gösta Holmér (D) Woldemar Gerschler [PYQ 2024] | Confidence: High
- Q11.** Which of the following methods involves a rapid, bouncing, and rhythmic movement to improve flexibility? (A) Slow stretching (B) Slow stretch and hold (C) Ballistic Method (D) PNF Technique [PYQ] | Confidence: Very High
- Q12.** What is the expanded form of PNF, a highly advanced technique to develop flexibility? (A) Proprioceptive Neural Facilitation (B) Proprioceptive Neuromuscular Facilitation (C) Passive

Neuromuscular Flexibility (D) Primary Neuromuscular Facilitation [PYQ] | Confidence: Very High

Q13. High pressure over muscles without changing the length of the muscle fibers is seen in which set of exercises? (A) Isometric (B) Isotonic (C) Isokinetic (D) Ballistic [PYQ] | Confidence: High

Q14. Given below are two statements labeled Assertion (A) and Reason (R): Assertion (A): Fartlek training is highly effective for developing both aerobic and anaerobic endurance. Reason (R): Fartlek training involves continuous running combined with intermittent speed bursts across varying natural terrains without a pre-planned strict pace. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [Practice] | Confidence: Very High

Q15. The coordinative ability that enables a sportsperson to accurately determine and swiftly change the position of their body in space is known as: (A) Coupling ability (B) Orientation ability (C) Reaction ability (D) Adaptation ability [PYQ] | Confidence: High

Q16. Define 'Endurance'. [PYQ 2019] | Confidence: Very High

Q17. Which component of physical fitness is most important for a combat sport like wrestling or judo? (A) Speed (B) Strength (C) Flexibility (D) Fine motor skills [PYQ] | Confidence: High

Q18. Which of the following exercises uses equipment to maintain a constant speed of movement regardless of the force applied? (A) Isometric (B) Isotonic (C) Isokinetic (D) Calisthenics [PYQ 2024] | Confidence: Very High

Q19. Given below are two statements labeled Assertion (A) and Reason (R): Assertion (A): Isotonic exercises are widely used by weightlifters to develop dynamic strength. Reason (R): In isotonic exercises, the length of the muscle changes (shortens and lengthens), and movement is clearly visible. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [Practice] | Confidence: High

Q20. A sprint runner quickly achieving their top speed from a stationary starting block is a demonstration of: (A) Locomotor ability (B) Acceleration ability (C) Reaction ability (D) Movement speed [Practice] | Confidence: High

2-Mark

Q21. Define 'Flexibility' and list down its two main types. [PYQ 2024] | Confidence: Very High

Q22. Briefly mention two advantages of Weight Training. [PYQ 2019] | Confidence: High

Q23. What do you understand by the term 'Talent Identification' in sports? [Practice] | Confidence: Very High

Q24. State the fundamental difference between Isometric and Isotonic exercises. [PYQ] | Confidence: Very High

- Q25.** Mention any two major advantages of the Fartlek training method. [PYQ] | Confidence: High
- Q26.** Differentiate between 'Active Flexibility' and 'Passive Flexibility'. [Practice] | Confidence: Very High
- Q27.** Explain the concept of 'Pace Runs' used for developing speed. [Practice] | Confidence: Medium
- Q28.** What is the 'Effort and Recovery' principle in the Interval Training method? [PYQ] | Confidence: High
- Q29.** Mention the names of any four types of coordinative abilities in sports. [PYQ] | Confidence: Very High
- Q30.** Define 'Strength'. Enlist its major types. [PYQ 2024] | Confidence: Very High
- Q31.** How does the 'Circuit Training' method benefit an athlete? State any two points. [Practice] | Confidence: High
- Q32.** Briefly explain the 'Reaction Ability' with a suitable sports example. [PYQ] | Confidence: Medium
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3-Mark

- Q33.** Define Speed and elaborate on 'Acceleration Runs' as a method to develop speed. [PYQ] | Confidence: Very High
- Q34.** Describe the Fartlek training method. How does it improve an athlete's endurance? [PYQ 2019] | Confidence: Very High
- Q35.** Explain the 'PNF' (Proprioceptive Neuromuscular Facilitation) technique. How is it executed to develop an athlete's flexibility? [PYQ] | Confidence: High
- Q36.** Define 'Sports Training'. Briefly explain the three periods of sports training cycles (Micro, Meso, and Macro). [Practice] | Confidence: Very High
- Q37.** Define strength and differentiate clearly between Isometric, Isotonic, and Isokinetic exercises. [PYQ 2024] | Confidence: Very High
- Q38.** Explain any three types of Coordinative Abilities that are essential for a gymnast. [PYQ] | Confidence: High
- Q39.** Define 'Endurance'. Elucidate the 'Continuous Training Method' and state its two advantages. [PYQ] | Confidence: Very High
- Q40.** Discuss the concepts of 'Talent Identification' and 'Talent Development'. Why are they crucial for a nation's sporting success? [Practice] | Confidence: High
- Q41.** Explain the 'Ballistic Method' and the 'Slow Stretch and Hold Method' for improving flexibility. [PYQ] | Confidence: Medium

Q42. Detail the procedure and utility of 'Interval Training' in building an athlete's cardiovascular endurance. [PYQ] | Confidence: High

5-Mark (Long Answer/Case-Study)

Q43. What do you understand by Circuit Training? How will a coach plan a circuit training session with 6 stations to develop the fitness of his new trainees? Explain with a diagram or flow. [PYQ 2025] | Confidence: Very High

Q44. Define 'Coordinative Abilities' as a crucial component of sports training. Thoroughly elaborate on any four types of coordinative abilities (such as Orientation ability, Coupling ability, Reaction ability, Balance ability) giving relevant sports examples. [PYQ] | Confidence: Very High

Q45. Define 'Flexibility'. Explain the various methods used in sports training to improve flexibility (Slow Stretching, Ballistic, PNF) in detail. [PYQ] | Confidence: Very High

Q46. Define 'Endurance'. Elucidate any two specific methods of developing endurance (Continuous, Interval, or Fartlek method) in detail, highlighting their procedures and physiological benefits. [PYQ] | Confidence: Very High

Q47. Case Study: Mr. Gopichand is a renowned badminton coach. When he started his academy, he selected school badminton players and designed a training program. During the training, he noted that a few players were good in defense but due to a lack of endurance and strength, they were unable to play up to the last moment. He used various methods to enhance their endurance and strength. (i) This type of training and exercises help in increasing static strength and maximal strength. (A) Isometric (B) Isotonic (C) Isokinetic (D) Aerobic (ii) Isotonic exercise helps in enhancing __. (A) Speed (B) Strength (C) Agility (D) Endurance (iii) High pressure over muscles can be seen in this set of exercises: (A) Isometric (B) Isotonic (C) Both A & B (D) Ballistic (iv) Which endurance training method involves periods of hard work followed by a time period of rest repeatedly? (A) Interval (B) Continuous (C) Fartlek (D) Pace runs [PYQ] | Confidence: Very High

Q48. Case Study: Sunita, a student of class VIII, was identified as a strong girl both physically and mentally. She is being encouraged by her teacher to take up wrestling as a professional sport and start training. Sunita is also interested in the sport. Her father, seeing her interest, sent her to a professional coach to learn the sport properly. (i) Which component of physical fitness is most important for a combat sport like wrestling? (A) Speed (B) Strength (C) Endurance (D) Flexibility (ii) Which type of body type is most suited for heavy wrestling? (A) Ectomorph (B) Mesomorph (C) Micromorph (D) Endomorph (iii) Which type of training method uses machines to regulate movement speed to develop strength? (A) Interval Training (B) Isometric Training (C) Isokinetic Training (D) Acceleration runs (iv) To hold an opponent down steadily, Sunita will mostly use which type of muscular contraction? (A) Isotonic (B) Isometric (C) Ballistic (D) Plyometric [PYQ] | Confidence: High

Q49. Case Study: Ravi has the aim of joining a uniform service like the police or army. But he has not qualified the **1500m** run in their selection criteria. Without qualifying this run, he cannot go to the next level. His coach designs a 6-week training block for him. (i) The **1500m** run is conducted to find the: (A) Endurance ability (B) Speed (C) Strength (D) Explosive power (ii) The best training method for the development of endurance by running across natural terrain with varying speeds is: (A) Continuous training (B) Interval training (C) Circuit training (D) Fartlek training (iii) The Interval training method is based on the principle of: (A) Overload (B) Effort (C) Effort and Recovery (D) Complete Rest (iv) The **6-week** training block Ravi's coach designed is scientifically known as a ___ cycle. [PYQ] | Confidence: Very High

Q50. Define 'Sports Training'. Discuss the concepts of Talent Identification and Talent Development. Furthermore, explain the specific physiological methods (Isometric, Isotonic, Isokinetic) used to develop an athlete's muscular strength. [Practice] | Confidence: High

Answer Key

Q1. (A) Fartlek training **Q2.** (C) Iso-metric **Q3.** (C) Length **Q4.** (B) Fartlek Training **Q5.** (C) Effort and Recovery **Q6.** (A) Endurance ability **Q7.** (B) Acceleration Ability (This is a type of speed, not coordinative ability). **Q8.** (B) Meso cycle **Q9.** (C) Macro cycle **Q10.** (A) J.J. Perrine **Q11.** (C) Ballistic Method **Q12.** (B) Proprioceptive Neuromuscular Facilitation **Q13.** (A) Isometric **Q14.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q15.** (B) Orientation ability **Q16.** Endurance is the ability of an organism to exert itself and remain active for a long period of time, resisting fatigue. **Q17.** (B) Strength **Q18.** (C) Isokinetic **Q19.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q20.** (B) Acceleration ability **Q21.** Flexibility is the maximum range of motion around a joint. Types: 1. Active Flexibility 2. Passive Flexibility. **Q22.** 1. Increases bone density and muscle mass (hypertrophy). 2. Improves joint strength and prevents sports injuries. **Q23.** Talent identification is the process of recognizing current participants with the potential to excel in a particular sport using scientific, physiological, and psychological testing. **Q24.** In Isometric exercises, muscle length remains constant and movement is not visible (e.g., pushing a wall). In Isotonic exercises, muscle length changes (concentric/eccentric) and movement is clearly visible (e.g., lifting a dumbbell). **Q25.** 1. Develops both aerobic and anaerobic endurance. 2. Not rigid; being in nature avoids boredom and psychological fatigue. **Q26.** Active flexibility is the ability to do movements with a greater amplitude without external help (e.g., stretching on your own). Passive flexibility uses external help (a partner or equipment) to achieve a greater stretch. **Q27.** Pace runs involve running a specific distance (usually **800m** or more) at a steady, unvarying, and pre-determined speed. It builds speed endurance. **Q28.** The athlete exercises at high intensity (Effort), causing an elevated heart rate, followed by a period of low intensity/walking (Recovery) before the heart rate fully returns to normal, then repeats. **Q29.** 1. Orientation ability. 2. Coupling ability. 3. Reaction ability. 4. Balance ability. **Q30.** Strength is the ability of a muscle to overcome resistance. Types: Maximum Strength, Explosive Strength, and Strength Endurance. **Q31.** 1. Enhances both muscular strength and cardiovascular endurance

simultaneously. 2. Accommodates multiple athletes at once in a small space with varying stations. **Q32.** Reaction ability is the ability to react quickly and effectively to a signal. Example: A sprinter exploding out of the starting blocks the exact moment the starter gun fires. **Q33.** Speed is the ability to perform a movement in the shortest possible time. Acceleration Runs are used to develop speed; the athlete starts from a stationary position and attains maximum speed as quickly as possible over a short distance (30m-50m), repeating it after full recovery. **Q34.** Fartlek means "Speed Play". It involves continuous running combined with interval training over natural, uneven terrain (hills, forests). The athlete changes pace (sprinting, jogging, walking) according to their own body's feeling. It boosts $\dot{V}O_2$ max, cardiovascular efficiency, and both aerobic/anaerobic endurance. **Q35.** PNF is an advanced flexibility method used in rehabilitation and sports. It involves stretching a muscle to its limit (passive stretch), then isometrically contracting it against a partner's resistance for a few seconds, relaxing, and then stretching it further. This suppresses the stretch reflex, allowing greater range of motion. **Q36.** Sports training is a scientific, planned process to improve sports performance. **Micro cycle:** Shortest training period (3-10 days), focusing on immediate skill/fitness goals. **Meso cycle:** Medium duration (3-6 weeks), consisting of several micro-cycles targeting specific fitness phases. **Macro cycle:** Longest period (3-12 months), aiming for peak performance at the main competition. **Q37.** Strength is the muscle's ability to exert force. **Isometric:** Iso=Constant, Metric=Length. Muscles exert force but do not change length; no visible movement (e.g., holding a plank). **Isotonic:** Iso=Constant, Tonic=Tension. Muscles shorten and lengthen, movement is visible (e.g., squats). **Isokinetic:** Kinetic=Motion. Speed of movement is kept strictly constant using special computerized machines throughout the range of motion. **Q38.** 1. **Balance Ability:** Maintaining body equilibrium during aerial or dynamic movements (e.g., landing from a vault). 2. **Orientation Ability:** Determining the body's position in space and time during complex twists. 3. **Coupling Ability:** Coordinating different body part movements seamlessly (e.g., combining a run, jump, and flip). **Q39.** Endurance is the ability to sustain an activity for a long duration, resisting fatigue. **Continuous Method:** Exercising for a long duration (30+ minutes) without any rest pauses at a steady heart rate (140-160 bpm). Advantages: Increases glycogen stores, improves heart/lung efficiency, and builds sheer willpower. **Q40.** **Talent Identification:** Scientifically screening young children to find those with the genetic, physical, and mental traits suited for a specific sport. **Talent Development:** Providing these identified individuals with world-class coaching, nutrition, and infrastructure to reach elite levels. Crucial for maximizing a nation's Olympic medal tally and avoiding wasted resources on unsuitable sports. **Q41.** **Ballistic Method:** Using momentum from a moving body or limb to force it beyond its normal range of motion; done rhythmically with bouncing/jerky movements. **Slow Stretch and Hold:** Slowly stretching a joint to its maximum limit and holding that exact position for 10-30 seconds. It is safer and prevents muscle tearing compared to the ballistic method. **Q42.** Interval Training is based on 'Effort and Recovery'. Procedure: Athlete runs a set distance (e.g., 400m) at 80% intensity, raising heart rate to 170-180 bpm. Then, they actively rest (walk/jog) until the heart rate drops to 120 bpm. Before it drops fully, the next run starts. It drastically improves the heart's stroke volume and lactic

acid tolerance. **Q43.** Circuit training is a series of **6 to 10** exercises (stations) performed one after another with minimal rest. It develops strength, endurance, and agility. **Coach's Plan for 6 Stations:** Station 1: Push-ups (Chest/Arms). Station 2: Skipping (Cardio). Station 3: Sit-ups (Core). Station 4: Squat Jumps (Legs). Station 5: Pull-ups (Back). Station 6: Shuttle run (Agility). *The athlete spends 30 seconds at each station, resting 15 seconds while rotating, completing 3 full circuits.* **Q44.** Coordinative abilities rely on the central nervous system to perform movements smoothly and accurately. 1. **Orientation:** Adjusting body position in space/time (e.g., a gymnast flipping). 2. **Coupling:** Coordinating multiple body parts together (e.g., a boxer coordinating footwork and a punch). 3. **Reaction:** Reacting quickly to a stimulus (e.g., goalkeeper saving a penalty). 4. **Balance:** Maintaining equilibrium in unstable postures (e.g., walking on a balance beam). **Q45.** Flexibility is the joint's maximum range of motion. **Methods:** 1. *Slow Stretching:* Gradually stretching to the max point and holding it. 2. *Ballistic:* Bouncing or swinging a limb rhythmically to force it past its normal range (e.g., dynamic leg swings). 3. *PNF (Proprioceptive Neuromuscular Facilitation):* Combining passive stretching and isometric contractions with a partner. The muscle is stretched, contracted against resistance, relaxed, and stretched further. It is the most effective method for elite athletes. **Q46.** Endurance is the ability to resist fatigue. 1. **Continuous Training:** Exercising without a break for **30+** minutes (e.g., cross-country running). Heart rate stays around **140-160** bpm. It heavily boosts aerobic capacity (VO_2 max) and burns fat. 2. **Fartlek Training:** 'Speed Play'. Running continuously but varying the speed (sprint, jog, walk) based on natural terrain (hills, sand). It develops both aerobic and anaerobic endurance and prevents mental boredom since the pace is self-regulated. **Q47.** (i) (A) Isometric. (ii) (B) Strength. (iii) (A) Isometric. (iv) (A) Interval. **Q48.** (i) (B) Strength. (ii) (B) Mesomorph (Muscular build). (iii) (C) Isokinetic Training. (iv) (B) Isometric (constant tension without movement). **Q49.** (i) (A) Endurance ability. (ii) (D) Fartlek training. (iii) (C) Effort and Recovery. (iv) Meso (Medium duration cycle of 3-6 weeks). **Q50.** Sports Training is the pedagogical process of athletic preparation. **Talent ID & Dev:** Finding genetically gifted youths and grooming them scientifically for elite performance. **Muscular Strength Methods:** 1. *Isometric:* Contraction against an immovable object (pushing a wall); builds static strength. 2. *Isotonic:* Contraction with visible joint movement (lifting weights); builds dynamic strength. 3. *Isokinetic:* Contraction at a constant speed using computerized machines (Cybex machines); maximizes force application safely for rehabilitation and elite sports.